The Farmer, Goose, Fox and Grain Problem

- Learn how to perform abstractions in order to make the problem solvable using Search
- States are generated on-the-fly; that is, neither the graph, nor the tree is given beforehand

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How to approach this problem using AI?

The farmer-fox-goose-grain problem.

A farmer has to cross a river with his fox, goose and grain. However, his little boat can only accommodate himself and one of his possessions. Therefore, in order to bring all of them across the river, he needs to cross it several times. During planning, he is cautious of two unwanted situations: an unguarded fox will eat the goose, and an unguarded goose will eat the grain.



Image: http://cairnarvon.rotahall.org/2010/02/28/search-trees-are-easy/

How can he cross the river without losing his possessions?

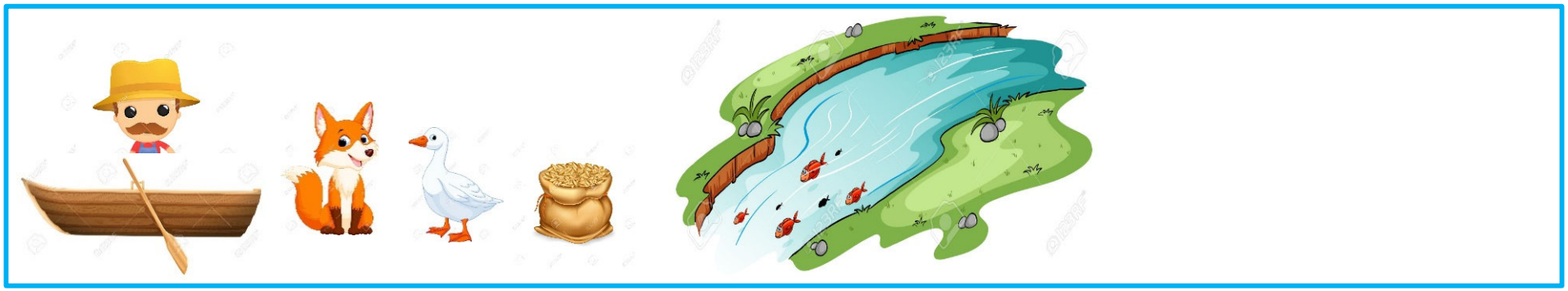
How to represent the problem as a Search problem?

What are the constraints to consider in this problem?

How many possible arrangements (states) are there? How many valid states?

Preconditions

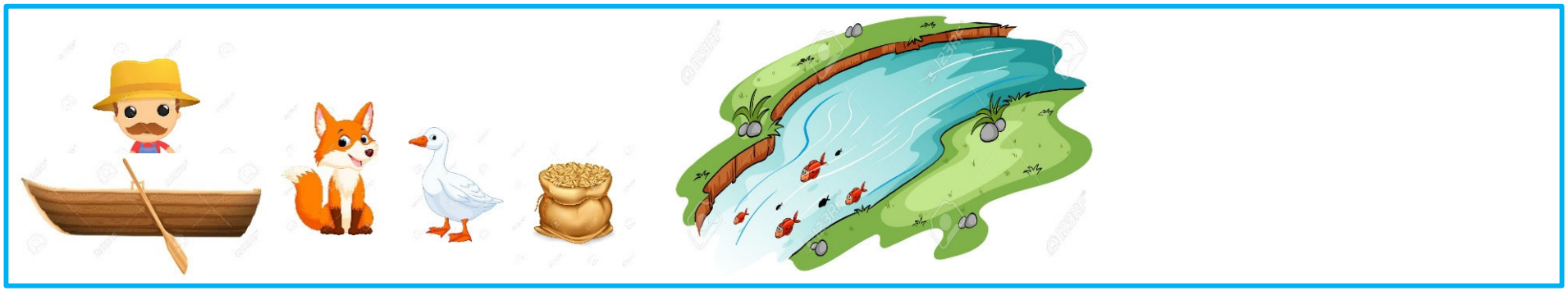
- The **fox** and the **goose** should not be in the same side of the river, not unless the farmer is with them.
- The **goose** and the **grain** should not be in the same side of the river, not unless the farmer is with them.



Left river bank river right river bank

Rules

1. Farmer can travel alone to the right river bank.
2. Farmer can travel alone to the left river bank.
3. Farmer can bring one of his possessions to the right river bank, provided that all preconditions are met.
4. Farmer can bring one of his possessions to the left river bank, provided that all preconditions are met.
5. The farmer should cross the river, every time he makes a move.



Left river bank river right river bank

Representation

How to represent the problem as a Search problem?



The objective is to move all of the farmer's possessions to the right side of the river bank.

What are the **States? Actions? Goal test?**

Initial state of the problem:





Left river bank

river

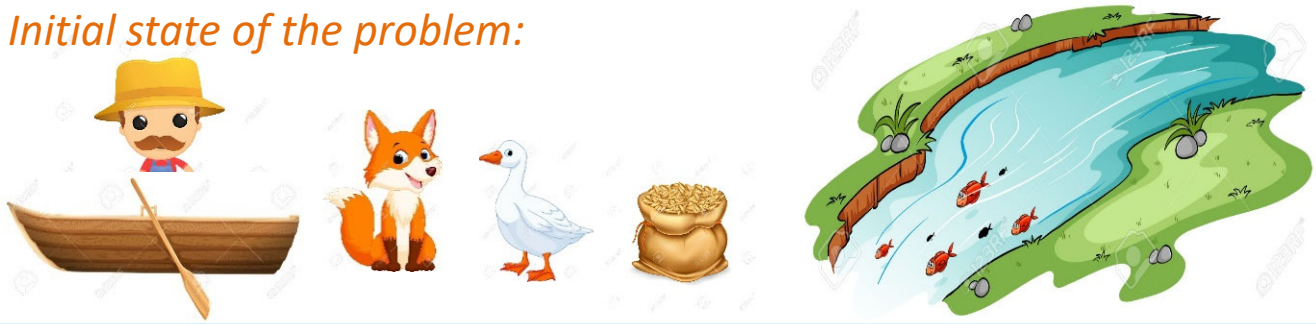
right river bank

States

The **state** is a tuple of **4 booleans**, representing the position of the farmer, goose, fox and grain, relative to the river.

Fa, Fo, Go, Gr

Initial state of the problem:



Left river bank river right river bank

Notation: When all 4 boolean variables are set to false, that means the farmer and all of his possessions are in the left side of the river bank.

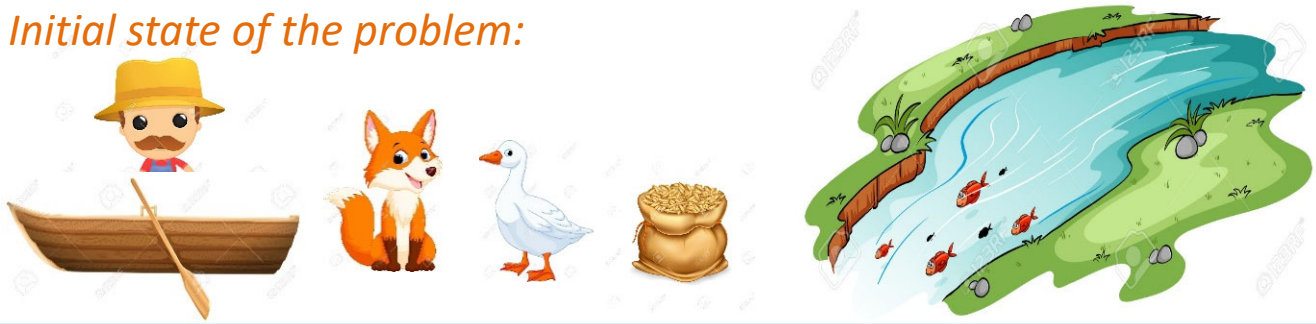
Fa=false, Fo=false, Go=false, Gr=false → 0, 0, 0, 0

States

The **state** is a tuple of **4 booleans**, representing the position of the farmer, goose, fox and grain, relative to the river.

Fa, Fo, Go, Gr

Initial state of the problem:



Left river bank river right river bank

Notation: When all 4 boolean variables are set to false, that means the farmer and all of his possessions are in the left side of the river bank.

Fa=false, Fo=false, Go=false, Gr=false → 0, 0, 0, 0

In this hand-simulation, to make the notations more intuitive, we can write the same state as follows:

< Fa, Fo, Go, Gr | >

The bar indicates the position of the river, relative to all characters.

Actions/Operators

There are **4 operators:** the farmer crosses alone, or with one of his three possessions. In each case all the preconditions must be met.

e.g. the farmer travels to the right river bank, the precondition is goose not equal fox and goose not equal grain.



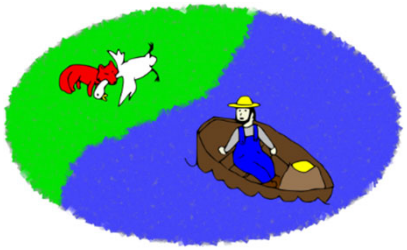
Actions/Operators/Moves:

1. Farmer travels alone to the left river bank.
2. Farmer travels alone to the right river bank.
3. Farmer travels with one of his possessions to the left river bank.
4. Farmer travels with one of his possessions to the right river bank.

Invalid States

Actions/Operators/Moves:

1. Farmer travels alone to the left river bank.
2. Farmer travels alone to the right river bank.
3. Farmer travels with one of his possessions to the left river bank.
4. Farmer travels with one of his possessions to the right river bank.



The following are invalid states that we should avoid:

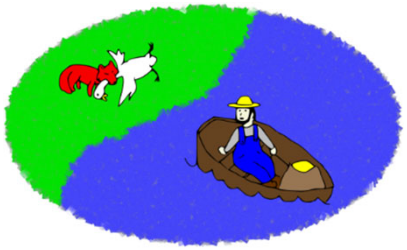
When the farmer travels alone to one of the river banks:

Fo, Go, Gr | **Fa**	The goose eats the grain, or the fox eats the goose first.
Fa | Fo, Go, Gr	The goose eats the grain, or the fox eats the goose first.

Invalid States

Actions/Operators/Moves:

1. Farmer travels alone to the left river bank.
2. Farmer travels alone to the right river bank.
3. Farmer travels with one of his possessions to the left river bank.
4. Farmer travels with one of his possessions to the right river bank.



The following are invalid states that we should avoid:

When the farmer travels alone to one of the river banks:

Fo, Go, Gr | **Fa**	The goose eats the grain, or the fox eats the goose first.
Fa | Fo, Go, Gr	The goose eats the grain, or the fox eats the goose first.

When the farmer travels with one of his possessions to one of the river banks:

Fo, Go | **Fa**, Gr	the fox eats the goose.
Fa, Gr | Fo, Go	the fox eats the goose.
Go, Gr | **Fa**, Fo	the goose eats the grain.
Fa, Fo | Go, Gr	the goose eats the grain.

There are (2^4)=**16 states** in total, but **6 are invalid**. Therefore, there are only **10 valid** states.

Goal Test

How to check if the algorithm has succeeded already?



Goal test:

All Booleans are true. That means all possessions, including the farmer are already on the right river bank.

i.e. farmer (**Fa**) == fox (**Fo**) == goose (**Go**) == grain (**Gr**) == true

Using our notation for the hand-simulation:

| Fa, Fo, Go, Gr

*To keep track of the characters in the puzzle more easily, enumerate them in alphabetical order

Exercise

Solve the problem using Depth-First Search

In your solution, consider the following sequence of moves:

1. Farmer travels alone to the other side of the river.
2. Farmer takes fox to the other side of the river.
3. Farmer takes goose to the other side of the river.
4. Farmer takes grain to the other side of the river.

Initial state of the problem:





Left river bank

river

right river bank